

Differential-Flatness-Based Feedforward and Speed-Feedback Control of a Salient-Pole Permanent Magnet Synchronous Motor

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Abstract

This paper develops a differential-flatness-based feedforward and speed-feedback controller for a salient-pole permanent magnet synchronous motor (PMSM), explicitly retaining the distinction between the direct-axis and quadrature-axis inductances, $L_d \neq L_q$.

The proposed architecture combines a model-based differential flatness feedforward component with a single mechanical speed-feedback term. A jerk-limited reference trajectory provides the desired rotor speed and its first two derivatives, ω^* , $\dot{\omega}^*$, and $\ddot{\omega}^*$. The mechanical dynamics are then inverted to determine the electromagnetic torque and q -axis current required to follow the desired trajectory.

The actual tracking error is defined as

$$e_\omega = \omega^* - \omega. \quad (1)$$

A desired first-order error dynamics,

$$\dot{e}_\omega = -k_\omega e_\omega, \quad k_\omega > 0, \quad (2)$$

is imposed directly on the mechanical system. This results in the commanded torque

$$T_e^{\text{cmd}} = J [\dot{\omega}^* + k_\omega(\omega^* - \omega)] + B\omega + T_L \quad (3)$$

where J is the mechanical inertia, B is the viscous friction coefficient, and T_L is the load torque.

The salient PMSM electromagnetic model is retained throughout the derivation. In particular, the electromagnetic torque contains the reluctance component associated with $L_d - L_q$. The commonly used operating condition $i_d^* = 0$ is subsequently introduced, under which the nominal torque mapping reduces to a linear relationship between q -axis current and electromagnetic torque.

The resulting controller provides a mathematically transparent separation between trajectory-based feedforward control and speed-error feedback. Stability of the ideal mechanical error dynamics is established using a Lyapunov function. Practical constraints on speed, acceleration, jerk, current, torque, and inverter voltage are also formulated.

1 Introduction

Differential flatness provides a systematic method for constructing feedforward control inputs from a desired trajectory and a mathematical model of a nonlinear system. Instead of relying exclusively on instantaneous tracking error, the desired trajectory and its derivatives are used to reconstruct the states and control inputs required by the plant.

Permanent magnet synchronous motors are nonlinear electromechanical systems in which the mechanical dynamics are directly related to electromagnetic torque, while the electromagnetic torque is related to the d - and q -axis stator currents.

For a salient PMSM, the distinction

$$\boxed{L_d \neq L_q} \quad (4)$$

must be retained because the difference between the two inductances produces a reluctance-torque contribution.

The objective of this work is to develop a simple real-time control architecture based on

$$\boxed{\text{Differential-flatness feedforward + speed-error feedback.}} \quad (5)$$

The controller does not introduce a conventional PI speed controller in the basic formulation. Instead, the speed feedback term is derived by explicitly selecting the desired dynamics of the speed tracking error.

The primary control error is

$$\boxed{e_\omega = \omega^* - \omega.} \quad (6)$$

The proposed architecture is therefore fundamentally different from a controller in which arbitrary proportional or PI gains are introduced without deriving their relationship to the PMSM mechanical dynamics.

The development proceeds in the following sequence:

1. derive the salient-PMSM model;
2. identify the flat-output trajectory;
3. derive the mechanical differential-flatness mapping;
4. introduce the speed tracking error;
5. impose stable error dynamics;
6. derive the commanded torque;
7. map torque to q -axis current;
8. derive the dq -axis feedforward voltages;
9. formulate physical constraints;
10. establish ideal closed-loop stability.

2 Salient PMSM Mathematical Model

Consider a three-phase salient-pole PMSM represented in the rotor-oriented dq reference frame.

The electrical equations are written as

$$\boxed{v_d = R_s i_d + L_d \dot{i}_d - p\omega L_q i_q} \quad (7)$$

and

$$\boxed{v_q = R_s i_q + L_q \dot{i}_q + p\omega (L_d i_d + \lambda_{PM}).} \quad (8)$$

Here,

$$v_d, v_q : \quad d\text{- and } q\text{-axis stator voltages,} \quad (9)$$

$$i_d, i_q : \quad d\text{- and } q\text{-axis stator currents,} \quad (10)$$

$$R_s : \quad \text{stator resistance,} \quad (11)$$

$$L_d : \quad d\text{-axis inductance,} \quad (12)$$

$$L_q : \quad q\text{-axis inductance,} \quad (13)$$

$$\lambda_{PM} : \quad \text{permanent-magnet flux linkage,} \quad (14)$$

$$p : \quad \text{number of pole pairs,} \quad (15)$$

$$\omega : \quad \text{mechanical rotor angular velocity.} \quad (16)$$

The salient-PMSM electromagnetic torque is

$$T_e = \frac{3}{2}p [\lambda_{PM}i_q + (L_d - L_q)i_di_q]. \quad (17)$$

Equation (17) contains two contributions:

$$T_e = T_{PM} + T_{rel}, \quad (18)$$

where

$$T_{PM} = \frac{3}{2}p\lambda_{PM}i_q \quad (19)$$

is the permanent-magnet torque and

$$T_{rel} = \frac{3}{2}p(L_d - L_q)i_di_q \quad (20)$$

is the reluctance torque.

Thus, unlike a non-salient PMSM,

$$L_d = L_q, \quad (21)$$

the salient machine satisfies

$$L_d \neq L_q. \quad (22)$$

The mechanical dynamics are

$$J\dot{\omega} = T_e - B\omega - T_L. \quad (23)$$

The mechanical position is related to speed by

$$\dot{\theta} = \omega. \quad (24)$$

The electrical rotor angle is

$$\theta_e = p\theta. \quad (25)$$

3 Flat Output and Reference Trajectory

The mechanical speed is selected as the principal flat output,

$$\boxed{y = \omega.} \quad (26)$$

The desired trajectory is represented by

$$\boxed{\omega^*, \quad \dot{\omega}^*, \quad \ddot{\omega}^*} \quad (27)$$

These quantities represent the desired angular velocity, angular acceleration, and angular jerk:

$$\omega^* = \text{desired angular velocity}, \quad (28)$$

$$\dot{\omega}^* = \text{desired angular acceleration}, \quad (29)$$

$$\ddot{\omega}^* = \text{desired angular jerk}. \quad (30)$$

A jerk-limited trajectory generator is used so that

$$\boxed{|\omega^*| \leq \omega_{\max}} \quad (31)$$

$$\boxed{|\dot{\omega}^*| \leq a_{\max}} \quad (32)$$

and

$$\boxed{|\ddot{\omega}^*| \leq j_{\max}.} \quad (33)$$

The use of a trajectory with bounded derivatives is important because the differential-flatness mapping requires the desired derivatives of the flat output.

4 Mechanical Differential-Flatness Mapping

Starting from the mechanical equation

$$J\dot{\omega} = T_e - B\omega - T_L, \quad (34)$$

the electromagnetic torque can be reconstructed as

$$\boxed{T_e = J\dot{\omega} + B\omega + T_L.} \quad (35)$$

For the desired trajectory,

$$\omega = \omega^*, \quad \dot{\omega} = \dot{\omega}^*, \quad (36)$$

the nominal feedforward torque becomes

$$\boxed{T_e^{FF} = J\dot{\omega}^* + B\omega^* + T_L.} \quad (37)$$

This is the mechanical differential-flatness feedforward mapping.

5 General Salient-PMSM Current Mapping

For the general salient PMSM, the torque equation is

$$T_e = \frac{3}{2}p [\lambda_{PM}i_q + (L_d - L_q)i_d i_q]. \quad (38)$$

Factoring i_q gives

$$T_e = \frac{3}{2}p [\lambda_{PM} + (L_d - L_q)i_d] i_q. \quad (39)$$

Consequently, the general q -axis current required to generate a specified electromagnetic torque is

$$i_q = \frac{T_e}{\frac{3}{2}p [\lambda_{PM} + (L_d - L_q)i_d]}. \quad (40)$$

Equation (40) explicitly demonstrates that for a salient PMSM the torque-to-current mapping depends on the operating d -axis current.

This is the general salient-machine relationship.

6 Zero- d -Axis-Current Operating Condition

For the baseline controller, the desired direct-axis current is chosen as

$$i_d^* = 0. \quad (41)$$

Consequently,

$$\dot{i}_d^* = 0. \quad (42)$$

Under this operating condition, the reluctance torque contribution evaluated on the nominal trajectory is zero:

$$T_{rel}^* = \frac{3}{2}p(L_d - L_q)i_d^* i_q^* = 0. \quad (43)$$

The nominal electromagnetic torque therefore becomes

$$T_e^* = \frac{3}{2}p\lambda_{PM}i_q^*. \quad (44)$$

Define

$$K_T = \frac{3}{2}p\lambda_{PM}. \quad (45)$$

Then

$$T_e^* = K_T i_q^*. \quad (46)$$

The feedforward current is consequently

$$i_q^{FF} = \frac{J\dot{\omega}^* + B\omega^* + T_L}{K_T}. \quad (47)$$

Although the torque-current mapping in the selected operating condition does not explicitly contain $L_d - L_q$, the electrical voltage equations continue to contain both L_d and L_q . Therefore the salient-machine model must not be replaced by a model having a single inductance.

7 Derivative of the Feedforward Current

Differentiating equation (47) gives

$$\dot{i}_q^{FF} = \frac{J\ddot{\omega}^* + B\dot{\omega}^* + \dot{T}_L}{K_T}. \quad (48)$$

If the load torque is assumed constant,

$$\boxed{\dot{T}_L = 0}, \quad (49)$$

then

$$\boxed{\dot{i}_q^{FF} = \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T}}. \quad (50)$$

This relationship demonstrates the direct connection between the mechanical trajectory jerk and the required electrical current derivative.

8 Actual Speed Tracking Error

The actual control error is defined by

$$\boxed{e_\omega = \omega^* - \omega}. \quad (51)$$

Differentiation gives

$$\dot{e}_\omega = \dot{\omega}^* - \dot{\omega}. \quad (52)$$

The control objective is to force

$$e_\omega \rightarrow 0. \quad (53)$$

Instead of introducing a conventional PI speed controller, a desired first-order error dynamics is selected:

$$\boxed{\dot{e}_\omega = -k_\omega e_\omega}, \quad (54)$$

where

$$\boxed{k_\omega > 0}. \quad (55)$$

Substituting the definition of the error into equation (54),

$$\dot{\omega}^* - \dot{\omega} = -k_\omega(\omega^* - \omega). \quad (56)$$

Therefore the desired actual acceleration is

$$\boxed{\dot{\omega}_{des} = \dot{\omega}^* + k_\omega(\omega^* - \omega)}. \quad (57)$$

This equation is the origin of the feedback term.

9 Derivation of the Speed-Feedback Torque

The mechanical equation can be rearranged to obtain the torque required to generate a desired acceleration:

$$T_e = J\dot{\omega}_{des} + B\omega + T_L. \quad (58)$$

Substituting equation (57) gives

$$T_e^{cmd} = J[\dot{\omega}^* + k_\omega(\omega^* - \omega)] + B\omega + T_L. \quad (59)$$

This is the central equation of the proposed controller.

It can be separated into feedforward and feedback components:

$$T_e^{cmd} = T_e^{FF} + T_e^{FB}, \quad (60)$$

where

$$T_e^{FF} = J\dot{\omega}^* + B\omega + T_L \quad (61)$$

and

$$T_e^{FB} = Jk_\omega e_\omega. \quad (62)$$

Therefore,

$$T_e^{FB} = Jk_\omega(\omega^* - \omega). \quad (63)$$

The feedback term is thus directly proportional to the actual mechanical speed tracking error.

10 Alternative Separation Using Reference-Speed Feedforward

For analytical interpretation, the controller can alternatively be written as

$$\begin{aligned} T_e^{cmd} &= J\dot{\omega}^* + B\omega^* + T_L \\ &\quad + Jk_\omega(\omega^* - \omega) + B(\omega - \omega^*). \end{aligned} \quad (64)$$

Hence

$$T_e^{cmd} = T_e^{DF} + (Jk_\omega - B) e_\omega, \quad (65)$$

where

$$T_e^{DF} = J\dot{\omega}^* + B\omega^* + T_L. \quad (66)$$

This form shows that if the viscous friction term is evaluated at the measured speed in the implementation, the exact feedback contribution contains both the inertial error correction and the friction difference.

For implementation, equation (59) is preferred because it follows directly from the mechanical equation and avoids unnecessary algebraic manipulation.

11 Commanded q -Axis Current

Under the selected operating condition $i_d^* = 0$, the torque-current relationship is

$$T_e = K_T i_q. \quad (67)$$

Consequently,

$$i_q^{cmd} = \frac{T_e^{cmd}}{K_T}. \quad (68)$$

Substitution of equation (59) gives

$$i_q^{cmd} = \frac{J[\dot{\omega}^* + k_\omega(\omega^* - \omega)] + B\omega + T_L}{K_T}. \quad (69)$$

The current command can be decomposed as

$$i_q^{cmd} = i_q^{FF} + i_q^{FB}, \quad (70)$$

where

$$i_q^{FF} = \frac{J\dot{\omega}^* + B\omega + T_L}{K_T} \quad (71)$$

and

$$i_q^{FB} = \frac{Jk_\omega}{K_T} e_\omega. \quad (72)$$

Therefore the actual speed error is mapped directly into a corrective q -axis current.

12 Electrical Differential-Flatness Mapping

The salient PMSM d -axis equation is

$$v_d = R_s i_d + L_d \dot{i}_d - p\omega L_q i_q. \quad (73)$$

For the nominal trajectory,

$$i_d^* = 0, \quad \dot{i}_d^* = 0. \quad (74)$$

Therefore the nominal d -axis feedforward voltage is

$$v_d^{FF} = -p\omega^* L_q i_q^{cmd}. \quad (75)$$

Notice that the cross-coupling term contains L_q , not a generic inductance. The q -axis equation is

$$v_q = R_s i_q + L_q \dot{i}_q + p\omega (L_d i_d + \lambda_{PM}). \quad (76)$$

For

$$i_d^* = 0, \quad (77)$$

the nominal q -axis feedforward voltage becomes

$$\boxed{v_q^{FF} = R_s i_q^{cmd} + L_q \dot{i}_q^{FF} + p\omega^* \lambda_{PM}.} \quad (78)$$

Using equation (50),

$$\boxed{v_q^{FF} = R_s i_q^{cmd} + L_q \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T} + p\omega^* \lambda_{PM}.} \quad (79)$$

Thus, even though $i_d^* = 0$ removes the nominal reluctance torque, the electrical voltage equations explicitly retain

$$\boxed{L_d \neq L_q.} \quad (80)$$

13 General Salient-PMSM Voltage Mapping

For completeness, before imposing $i_d^* = 0$, the voltage equations corresponding to arbitrary flat trajectories are

$$\boxed{v_d^{FF} = R_s i_d^* + L_d \dot{i}_d^* - p\omega^* L_q i_q^*} \quad (81)$$

and

$$\boxed{v_q^{FF} = R_s i_q^* + L_q \dot{i}_q^* + p\omega^* (L_d i_d^* + \lambda_{PM}).} \quad (82)$$

The corresponding torque is

$$\boxed{T_e^* = \frac{3}{2} p [\lambda_{PM} i_q^* + (L_d - L_q) i_d^* i_q^*].} \quad (83)$$

These equations constitute the general salient-PMSM model inversion.

The zero- d -axis-current controller used in this work is obtained by the specialization

$$i_d^* = 0, \quad \dot{i}_d^* = 0. \quad (84)$$

14 Complete Differential-Flatness and Feedback Controller

The complete controller can now be written as

$$\boxed{\begin{aligned} e_\omega &= \omega^* - \omega, \\ \dot{\omega}_{des} &= \dot{\omega}^* + k_\omega e_\omega, \\ T_e^{cmd} &= J\dot{\omega}_{des} + B\omega + T_L, \\ K_T &= \frac{3}{2} p \lambda_{PM}, \\ i_q^{cmd} &= \frac{T_e^{cmd}}{K_T}, \\ i_d^* &= 0, \\ v_d^{cmd} &= -p\omega^* L_q i_q^{cmd}, \\ v_q^{cmd} &= R_s i_q^{cmd} + L_q \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T} + p\omega^* \lambda_{PM}. \end{aligned}} \quad (85)$$

The controller therefore has the structure

trajectory $\rightarrow (\omega^*, \dot{\omega}^*, \ddot{\omega}^*)$ $\rightarrow$ DF mechanical mapping $\rightarrow T_e^{FF}$ $+$ speed-error correction $\rightarrow T_e^{cmd}$ $\rightarrow i_q^{cmd}$ $\rightarrow (v_d^{cmd}, v_q^{cmd})$ $\rightarrow$ SVPWM.	(86)
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15 Closed-Loop Speed Error Dynamics

Consider the ideal mechanical PMSM equation

$$J\dot{\omega} = T_e - B\omega - T_L. \quad (87)$$

Apply the proposed controller

$$T_e^{cmd} = J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L. \quad (88)$$

Assuming ideal torque realization,

$$J\dot{\omega} = J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L - B\omega - T_L \quad (89)$$

$$= J[\dot{\omega}^* + k_\omega e_\omega]. \quad (90)$$

Consequently,

$\dot{\omega} = \dot{\omega}^* + k_\omega e_\omega.$
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(91)

Since

$$e_\omega = \omega^* - \omega, \quad (92)$$

its derivative is

$$\dot{e}_\omega = \dot{\omega}^* - \dot{\omega}. \quad (93)$$

Therefore,

$$\dot{e}_\omega = \dot{\omega}^* - [\dot{\omega}^* + k_\omega e_\omega] \quad (94)$$

$$= -k_\omega e_\omega. \quad (95)$$

Hence the closed-loop error dynamics are

$\dot{e}_\omega = -k_\omega e_\omega.$
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(96)

16 Exponential Convergence

The solution of equation (96) is

$$e_\omega(t) = e_\omega(0)e^{-k_\omega t}. \quad (97)$$

For

$$k_\omega > 0, \quad (98)$$

the tracking error converges exponentially to zero:

$$\lim_{t \rightarrow \infty} e_\omega(t) = 0. \quad (99)$$

The ideal mechanical time constant is

$$\tau_\omega = \frac{1}{k_\omega}. \quad (100)$$

Thus the feedback parameter has a direct physical interpretation.

17 Lyapunov Stability Analysis

Consider the positive-definite Lyapunov candidate

$$V(e_\omega) = \frac{1}{2}e_\omega^2. \quad (101)$$

For

$$e_\omega \neq 0, \quad (102)$$

we have

$$V > 0. \quad (103)$$

Differentiating,

$$\dot{V} = e_\omega \dot{e}_\omega. \quad (104)$$

Using the closed-loop error equation,

$$\dot{e}_\omega = -k_\omega e_\omega, \quad (105)$$

gives

$$\dot{V} = -k_\omega e_\omega^2. \quad (106)$$

For

$$k_\omega > 0, \quad (107)$$

we obtain

$$\dot{V} < 0 \quad \forall e_\omega \neq 0. \quad (108)$$

Therefore the ideal speed-tracking error is asymptotically stable. Because the error equation is linear and exponentially convergent, the equilibrium is globally exponentially stable within the domain in which the mathematical model and actuator commands remain valid.

18 Torque Constraint

A real PMSM has a finite current capability.

Let

$$|i_q^{cmd}| \leq I_{q,\max}. \quad (109)$$

For the selected $i_d^* = 0$ operating condition,

$$T_e = K_T i_q. \quad (110)$$

Therefore,

$$\boxed{|T_e^{cmd}| \leq T_{e,\max}} \quad (111)$$

where

$$\boxed{T_{e,\max} = K_T I_{q,\max}}. \quad (112)$$

The unconstrained speed-feedback controller therefore has to satisfy

$$\boxed{|J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L| \leq T_{e,\max}}. \quad (113)$$

This inequality defines a physical limitation on the allowable speed-tracking correction.

19 Current Constraint

From

$$i_q^{cmd} = \frac{J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L}{K_T}, \quad (114)$$

the current constraint is

$$\boxed{\left| \frac{J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L}{K_T} \right| \leq I_{q,\max}}. \quad (115)$$

Equivalently,

$$\boxed{|J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L| \leq K_T I_{q,\max}}. \quad (116)$$

20 Acceleration Constraint

The commanded mechanical acceleration is

$$\dot{\omega}_{des} = \dot{\omega}^* + k_\omega e_\omega. \quad (117)$$

If the motor is constrained to

$$|\dot{\omega}| \leq a_{\max}, \quad (118)$$

then the feedback-controlled acceleration must satisfy

$$\boxed{|\dot{\omega}^* + k_\omega e_\omega| \leq a_{\max}}. \quad (119)$$

This inequality is particularly important when a large speed error occurs during a rapid transient.

21 Jerk Constraint

The reference trajectory satisfies

$$\boxed{|\ddot{\omega}^*| \leq j_{\max}} \quad (120)$$

The nominal q -axis current derivative is

$$\dot{i}_q^{FF} = \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T}. \quad (121)$$

Therefore,

$$|\dot{i}_q^{FF}| \leq \frac{J|\ddot{\omega}^*| + B|\dot{\omega}^*|}{K_T}. \quad (122)$$

Using the trajectory limits,

$$\boxed{|\dot{i}_q^{FF}| \leq \frac{Jj_{\max} + Ba_{\max}}{K_T}} \quad (123)$$

This equation provides a direct relationship between the mechanical jerk constraint and the required electrical current slew rate.

22 Inverter Voltage Constraint

The inverter has a finite voltage capability. Let the available voltage magnitude be $V_{\max}$.

The voltage vector must satisfy

$$\boxed{v_d^2 + v_q^2 \leq V_{\max}^2} \quad (124)$$

For the proposed controller,

$$v_d^{cmd} = -p\omega^* L_q i_q^{cmd} \quad (125)$$

and

$$v_q^{cmd} = R_s i_q^{cmd} + L_q \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T} + p\omega^* \lambda_{PM}. \quad (126)$$

Consequently, the voltage constraint becomes

$$\boxed{\begin{aligned} & \left(-p\omega^* L_q i_q^{cmd}\right)^2 \\ & + \left(R_s i_q^{cmd} + L_q \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T} + p\omega^* \lambda_{PM}\right)^2 \leq V_{\max}^2. \end{aligned}} \quad (127)$$

This constraint is especially important at high rotor speed because the back-electromotive-force term

$$p\omega^* \lambda_{PM} \quad (128)$$

increases with speed.

23 Effect of Saliency

The use of

$$L_d \neq L_q \quad (129)$$

has two important consequences.

First, the general electromagnetic torque contains

$$T_{rel} = \frac{3}{2}p(L_d - L_q)i_d i_q. \quad (130)$$

Second, the electrical voltage equations contain different inductances:

$$v_d = R_s i_d + L_d \dot{i}_d - p\omega L_q i_q \quad (131)$$

and

$$v_q = R_s i_q + L_q \dot{i}_q + p\omega(L_d i_d + \lambda_{PM}). \quad (132)$$

Thus the machine cannot in general be represented by a single inductance.

For the baseline controller,

$$i_d^* = 0, \quad (133)$$

so the nominal reluctance torque is zero. Nevertheless, the distinction between L_d and L_q remains essential in the voltage mapping and in any future extension that uses nonzero i_d .

24 Generalization to Nonzero d -Axis Current

Although the present controller uses

$$i_d^* = 0, \quad (134)$$

the general salient-PMSM torque equation is

$$T_e = \frac{3}{2}p[\lambda_{PM} + (L_d - L_q)i_d]i_q. \quad (135)$$

Therefore, if a future controller selects

$$i_d^* \neq 0, \quad (136)$$

the required q -axis current becomes

$$i_q^* = \frac{T_e^*}{\frac{3}{2}p[\lambda_{PM} + (L_d - L_q)i_d^*]}. \quad (137)$$

This formulation allows the differential-flatness framework to be extended to maximum-torque-per-ampere operation, field weakening, or other optimal current trajectories.

The present paper deliberately does not introduce these additional control objectives so that the fundamental DF feedforward and speed feedback structure remains clear.

25 Discrete-Time Implementation

For a digital controller operating at

$$f_s = 20 \text{ kHz}, \quad (138)$$

the sampling interval is

$$T_s = \frac{1}{f_s} = 50 \text{ } \mu\text{s}. \quad (139)$$

At control step k , the measured speed is

$$\omega[k]. \quad (140)$$

The trajectory generator supplies

$$\omega^*[k], \quad \dot{\omega}^*[k], \quad \ddot{\omega}^*[k]. \quad (141)$$

The speed error is calculated as

$$e_\omega[k] = \omega^*[k] - \omega[k]. \quad (142)$$

The commanded torque is then

$$T_e^{cmd}[k] = J [\dot{\omega}^*[k] + k_\omega e_\omega[k]] + B\omega[k] + T_L. \quad (143)$$

The current command is

$$i_q^{cmd}[k] = \frac{T_e^{cmd}[k]}{K_T}. \quad (144)$$

The nominal current derivative is

$$\dot{i}_q^{FF}[k] = \frac{J\ddot{\omega}^*[k] + B\dot{\omega}^*[k]}{K_T}. \quad (145)$$

The voltage commands are

$$v_d^{cmd}[k] = -p\omega^*[k]L_q i_q^{cmd}[k] \quad (146)$$

and

$$v_q^{cmd}[k] = R_s i_q^{cmd}[k] + L_q \dot{i}_q^{FF}[k] + p\omega^*[k]\lambda_{PM}. \quad (147)$$

The resulting dq voltage vector is transformed to the stationary frame and applied through the inverter modulation stage.

26 Choice of the Speed Feedback Gain

The parameter k_ω determines the ideal convergence rate.

The corresponding first-order time constant is

$$\tau_\omega = \frac{1}{k_\omega}. \quad (148)$$

For example,

k_ω	τ_ω	Ideal interpretation
10 s ⁻¹	100 ms	Slow
20 s ⁻¹	50 ms	Moderate
50 s ⁻¹	20 ms	Fast
100 s ⁻¹	10 ms	Very fast

These values describe the ideal mathematical error dynamics only. The practical value must be selected according to the motor's available torque, current, voltage, mechanical inertia, load, and desired transient response.

In particular, increasing k_ω increases the feedback torque

$$T_e^{FB} = Jk_\omega e_\omega. \quad (149)$$

Therefore an excessively large value can cause torque saturation.

27 Actuator Saturation and Practical Stability

The exponential stability result

$$\dot{e}_\omega = -k_\omega e_\omega \quad (150)$$

assumes that the commanded torque and voltage can be realized by the physical actuator. In practice,

$$|i_q^{cmd}| \leq I_{q,\max} \quad (151)$$

and

$$v_d^2 + v_q^2 \leq V_{\max}^2. \quad (152)$$

If these constraints are violated, the actual torque differs from the ideal commanded torque:

$$T_e^{actual} \neq T_e^{cmd}. \quad (153)$$

Consequently, the ideal error dynamics no longer hold exactly.

The controller should therefore implement torque/current and voltage limiting before the final SVPWM stage.

The ideal stability proof should consequently be interpreted as the nominal closed-loop stability result inside the actuator-feasible region.

28 Control Architecture

The proposed controller can be summarized by the following sequence:

$$\boxed{\omega_{command} \rightarrow \text{jerk-limited trajectory} \rightarrow (\omega^*, \dot{\omega}^*, \ddot{\omega}^*)} \quad (154)$$

followed by

$$\boxed{e_\omega = \omega^* - \omega.} \quad (155)$$

The mechanical differential-flatness mapping produces

$$\boxed{T_e^{FF} = J\dot{\omega}^* + B\omega + T_L.} \quad (156)$$

The speed feedback produces

$$\boxed{T_e^{FB} = Jk_\omega e_\omega.} \quad (157)$$

The total torque command is

$$\boxed{T_e^{cmd} = T_e^{FF} + T_e^{FB}.} \quad (158)$$

The torque is then mapped to current:

$$\boxed{i_q^{cmd} = \frac{T_e^{cmd}}{K_T}.} \quad (159)$$

Finally, the salient-PMSM electrical model gives

$$\boxed{v_d^{cmd} = -p\omega^* L_q i_q^{cmd}} \quad (160)$$

and

$$\boxed{v_q^{cmd} = R_s i_q^{cmd} + L_q \dot{i}_q^{FF} + p\omega^* \lambda_{PM}.} \quad (161)$$

29 Summary of the Mathematical Controller

The entire proposed controller can be expressed compactly as

$$\boxed{\begin{aligned} e_\omega &= \omega^* - \omega, \\ T_e^{cmd} &= J[\dot{\omega}^* + k_\omega e_\omega] + B\omega + T_L, \\ K_T &= \frac{3}{2}p\lambda_{PM}, \\ i_q^{cmd} &= \frac{T_e^{cmd}}{K_T}, \\ i_d^* &= 0, \\ \dot{i}_q^{FF} &= \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_T}, \\ v_d^{cmd} &= -p\omega^* L_q i_q^{cmd}, \\ v_q^{cmd} &= R_s i_q^{cmd} + L_q \dot{i}_q^{FF} + p\omega^* \lambda_{PM}. \end{aligned}} \quad (162)$$

The principal stability condition is

$$\boxed{k_\omega > 0.} \quad (163)$$

The principal actuator constraints are

$$\boxed{|\omega^*| \leq \omega_{\max},} \quad (164)$$

$$\boxed{|\dot{\omega}^*| \leq a_{\max},} \quad (165)$$

$$\boxed{|\ddot{\omega}^*| \leq j_{\max},} \quad (166)$$

$$\boxed{|i_q^{cmd}| \leq I_{q,\max},} \quad (167)$$

and

$$\boxed{(v_d^{cmd})^2 + (v_q^{cmd})^2 \leq V_{\max}^2.} \quad (168)$$

30 Conclusion

A differential-flatness-based feedforward and speed-feedback controller has been derived for a salient-pole PMSM while explicitly retaining the general condition

$$\boxed{L_d \neq L_q.} \quad (169)$$

The complete salient-PMSM model includes the reluctance torque

$$T_{rel} = \frac{3}{2}p(L_d - L_q)i_d i_q \quad (170)$$

and the distinct d - and q -axis inductances in the voltage equations. For the baseline operating condition

$$i_d^* = 0, \quad (171)$$

the nominal reluctance torque vanishes and the torque-current relationship becomes

$$T_e^* = \frac{3}{2}p\lambda_{PM}i_q^*. \quad (172)$$

The desired mechanical trajectory is generated using

$$\omega^*, \quad \dot{\omega}^*, \quad \ddot{\omega}^*, \quad (173)$$

and the actual speed tracking error is

$$\boxed{e_\omega = \omega^* - \omega.} \quad (174)$$

By imposing the desired error dynamics

$$\boxed{\dot{e}_\omega = -k_\omega e_\omega,} \quad (175)$$

the commanded electromagnetic torque becomes

$$\boxed{T_e^{cmd} = J[\dot{\omega}^* + k_\omega(\omega^* - \omega)] + B\omega + T_L.} \quad (176)$$

This equation represents the central contribution of the proposed controller: the first term is the model-based trajectory feedforward, while the second term provides a direct correction based on the actual mechanical speed error.

The resulting ideal error dynamics are

$$\dot{e}_\omega = -k_\omega e_\omega, \quad (177)$$

with solution

$$e_\omega(t) = e_\omega(0)e^{-k_\omega t}. \quad (178)$$

A Lyapunov analysis using

$$V = \frac{1}{2}e_\omega^2 \quad (179)$$

gives

$$\boxed{\dot{V} = -k_\omega e_\omega^2 < 0} \quad (180)$$

for

$$k_\omega > 0. \tag{181}$$

Therefore the ideal speed tracking error converges exponentially to zero, provided the required torque and voltage remain within the physical capabilities of the drive.

The resulting architecture provides a simple mathematical foundation for real-time implementation of differential-flatness-based PMSM control. Future extensions can introduce nonzero i_d trajectories, maximum-torque-per-ampere operation, field weakening, load-torque observation, parameter adaptation, and explicit treatment of actuator saturation.